

SIGNALING SYSTEM 7 (SS7)

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WHAT IS COMMUNICATION ?

- Communication is used between 2 network elements to exchange information.
- There are 2 types of domains in the communication world
 - Data communication
 - Telecommunication

DIFFERENCES BETWEEN TELECOM & DATACOM

Telecommunication	Data Communication
Network used for making voice calls between telephones	Network used for transferring data from one computer to another
Telephones were the end points	Computers were the end point
Uses protocols like ISUP, ISDN	Predominantly uses TCP/IP
Network elements are MSC, HLR, SCP etc	Network elements are Routers, hubs, ATMs, bridges etc
Uses circuit switching	Uses packet switching

NETWORKS OF TODAY ...

- In the networks of today telecommunication networks are used to download data e.g.GPRS
- Data networks are used for making Voice Calls e.g. VOIP

WHAT IS SIGNALING ?

- Signaling refers to the exchange of information between network elements
- Signaling between network elements follows a specific protocol
- A Protocol refers to the set of rules for communicating between the elements

WHAT IS SS7 SIGNALING ?

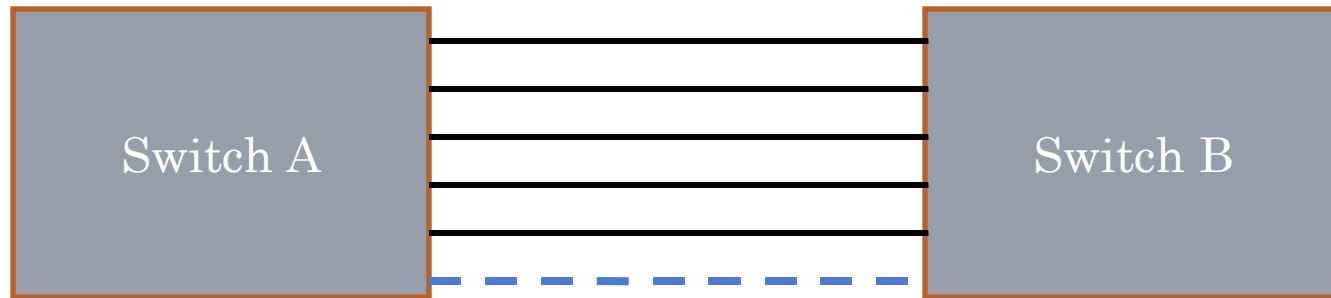
- Signaling System 7 was devised by ITU-T (formerly known as CCITT)
- Initially the signaling in trunks used to happen on the same channel in which the voice call used to occur. Hence call setup, digits etc would use the same line as the actual voice circuit
- In SS7 a separate channel is allocated just for signaling. This is known as Out-Of- Band signaling

SS7 SIGNALING

- Signaling happens in a separate channel outside of the voice channels
- A separate timeslot is used to transfer signaling messages like call setup, teardown etc
- This is also known as Common Channel Signaling (CCS)

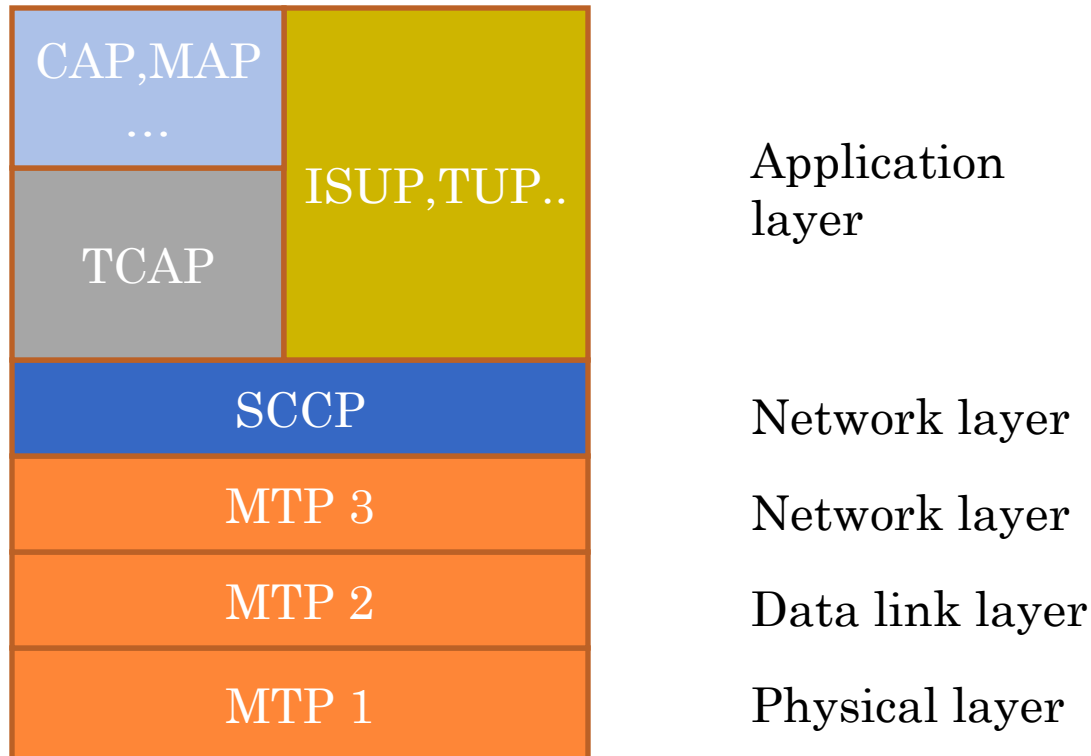
ADVANTAGES OF OUT OF BAND SIGNALING

- Allows signaling at any phase of the call
- Allows upto 56 Kbps of signaling information



— Voice Trunk
- - - Signaling Link

SS7 STACK

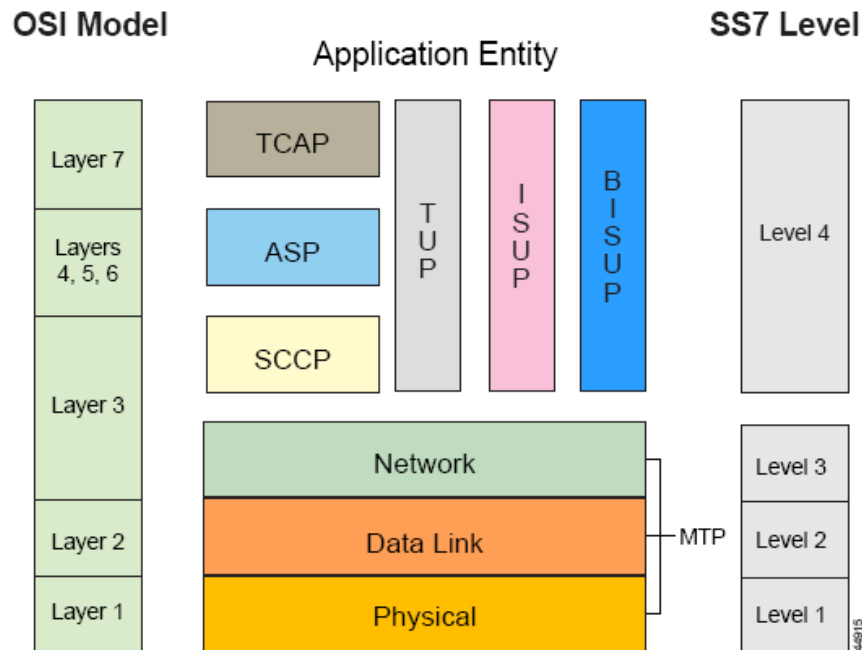


SS7 LAYERS

- MTP – Message Transfer Part
- SCCP – Signaling and Connection Control Part
- TCAP – Transaction Capabilities Application Part
- CAP – CAMEL Application Part
- ISUP – ISDN User Part
- MAP – Mobile Application Part

SS7 VS OSI STACK

Figure 3-1 SS7 Protocol Stack



TCAP: Transaction Capabilities Application Part

ASP: Application Service Part

SCCP: Signaling Connection Control Part

TUP: Telephone User Part

ISUP: ISDN User Part

BISUP: Broadband ISDN User Part

MTP: Message Transfer Part

SS7 PROTOCOLS

SS7 protocols can be classified as call related and non-call related

- Call related protocols – ISUP (ISDN User Part), B-ISUP (Broadband – ISUP)
- Non-call related protocols –
 - INAP (Intelligent Network Application Part)
 - CAP (CAMEL Application Part)
 - MAP (Mobile Application Part)
 - ...

SS7 STANDARD BODIES

- The SS7 standard body for North American market is ANSI (American National Standards Institute). The ANSI versions are ANSI ISUP, ANSI TCAP etc.
- The SS7 standard body for European market is ETSI (formerly CCITT). The ETSI versions of the protocols are ETSI ISUP, ETSI TCAP etc.
- There are minor variations in the protocols by the two standard bodies.

SS7 Link speeds

There are 2 main standards in the world for transmission of PCM signals

T-Carrier – This is used in the North American market

E-Carrier – This is used for European market

T1 – 24 channels * 64 Kbps = 1.544 Mbps

E1 – 32 channels * 64 Kbps = 2.048 Mbps

Level	North American	European
0	64 Kbps	64 kbps
1	1.544 Mbps (T1)	2.048 Mbps (E1)
2	6.312 Mbps (T2)	8.448 (E2)
3	44.736 Mbps (T3)	34.368 (E3)

MTP LAYER 1 – PHYSICAL LAYER

- This is the Physical Layer and defines the electrical and physical characteristics of SS7 links.
- Signaling links use DS-0 channels and carry data at 56 Kbps or 64Kbps.

MTP LAYER 2 – DATA LINK LAYER

- The data link level provides the network with sequenced delivery of all SS7 message packets between adjacent nodes
- It also provides for flow control and error checking

MTP LAYER 3 – NETWORK LAYER

- This provides network layer functionality. It delivers messages between signaling end points
- The network level depends on the services of Level 2 to provide alternate routing, and message distribution functions.

SCCP (SIGNALING CONNECTION CONTROL PART)

- Signaling Connection Control Part (SCCP) is a higher level protocol than MTP that provides end-to-end routing.
- SCCP is required for routing TCAP messages to their proper database.
- SCCP also does Global Title Translation to determine the end signaling point.

TCAP (TRANSACTION CAPABILITIES APPLICATION PART)

- Transactional Capabilities Application Part (TCAP) facilitates connection to an external database.
- Information/data received is sent back in the form of a TCAP message

APPLICATION LAYER

- At the application layer there are the following protocols
- Call related
 - ISDN User Part (ISUP) supports basic telephone call connect/disconnect between end offices.
- Non-call related
 - CAP – Camel Application Part is used to access a database ,the SCP and influence the call

SS7 SIGNALING ARCHITECTURE

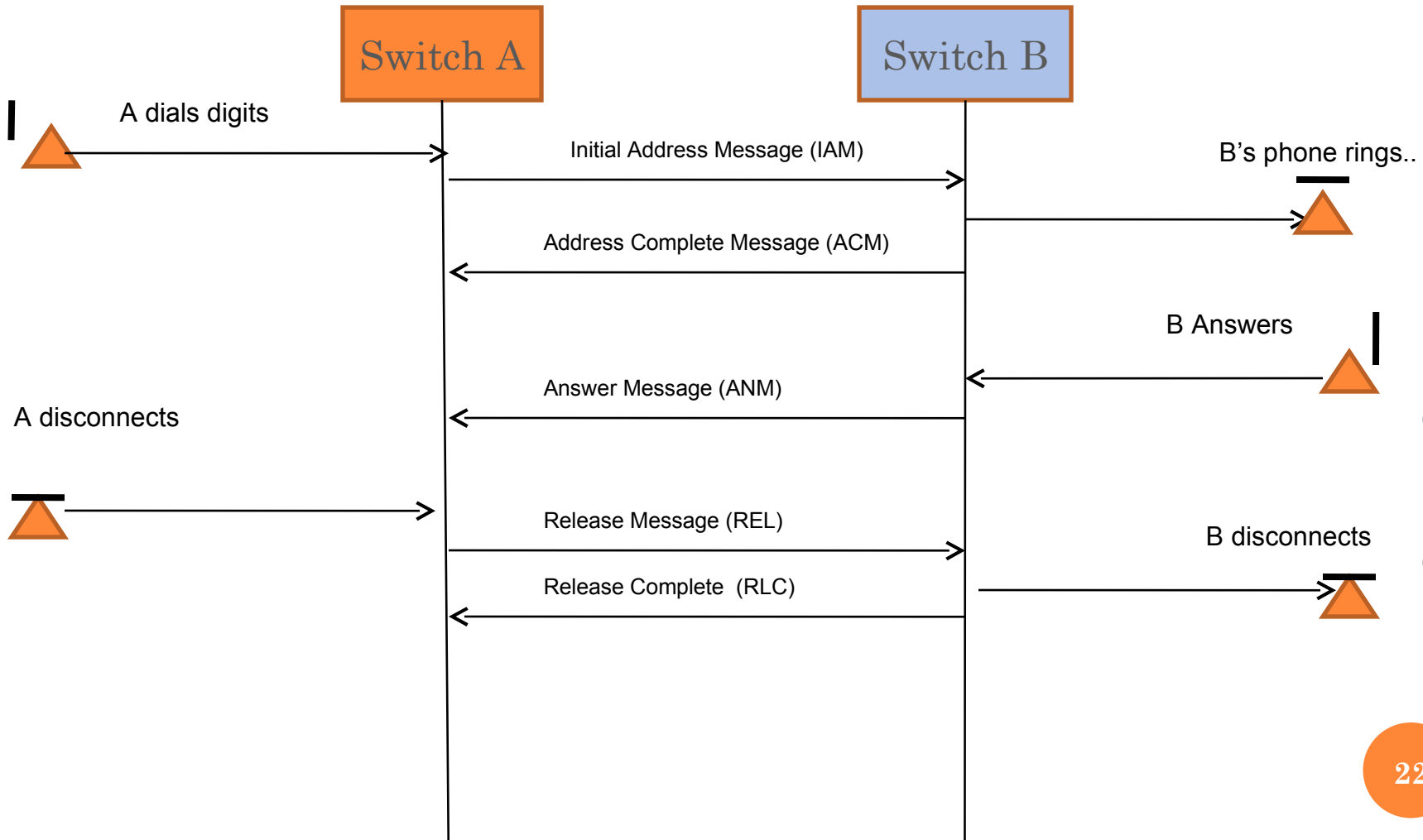
There are 3 main elements in SS7 signaling architecture

SSP – These are SS7 capable digital switches

STP – These are SS7 capable network elements that route incoming SS7 messages to the correct destination

SCP – These are databases which take part in non-call related SS7 signaling

ISUP CALL FLOW



Thank You !!!